MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT DIRECTOR (FOOD & DRUGS) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (NOVEMBER, 2025)

TECHNICAL PAPER-I

Time Allowed : 3 hours FM : 200

SECTION - A *(Conventional Type) (100 Marks)*

This Section should be answered only on the Answer Sheet provided.
Marks for each question is indicated against it.

1. Explain the relationship between a drug's pKa and its pH-dependent solubility. Describe the methods used for measuring tonicity. **(10)**

2. Discuss the types of excipients used in tablet formulation, including diluents, binders, disintegrants, and lubricants. Also discuss the importance of content uniformity and weight variation tests for tablets. **(10)**

3. What are the main components of a UV visible spectrophotometer? What is the effect of solvent on the absorption of UV visible spectroscopy? **(10)**

4. What are the various sources of impurities in pharmaceutical substances? How can impurities be minimized in pharmaceutical production? Why is it important to control impurities in pharmaceuticals? **(10)**

5. Explain the G- Protein – coupled receptors and transmembrane enzyme link receptors in detail with examples. **(10)**

6. Explain in detail the different phases for clinical evaluation of new drug molecules. **(10)**

7. What is the principle of Bioassay? What are the different types of bioassay with their specific applications? **(10)**

8. Explain NDA and ANDA in drug approval process. Explain and differentiate adverse drug events (ADEs) and adverse drug reactions (ADRs). **(6+4=10)**

9. Discuss the pharmacopoeial tests for quality control of tablet dosage form. **(10)**

10. How will you differentiate calibration from validation? Explain chromophores and auxochromes in the context of UV-Vis spectroscopy, providing suitable examples. **(6+4=10)**

SECTION - B *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.
*This Section should be answered only on the **OMR Response Sheet** provided.*

1. Which of the following is not an important parameter of plasma level time studies?
(a) Cmax (b) Tmax
(c) The area under the plasma level-time curve (d) Steady state level

2. A drug with poor stability means higher bioavailability.
(a) True
(b) False
(c) None
(d) All of the above

3. When rate is independent of the reactant concentration, it is called:
(a) Zero order reaction. (b) Pseudo zero order reaction
(c) First order reaction (d) Second order reaction

4. Which of the following is the half life of first order reaction?
(a) t1/2 = A0/2k (b) t1/2 = 0.693/2k
(c) t1/2 = 2k (d) t1/2 = 0.693/k

5. On a product, the label states 'protect from light'. What type of decomposition does the product undergo?
(a) Carboxylation (b) Decarboxylation
(c) Hydrolysis (d) Oxidation

6. Aspirin undergoes decomposition in a formulation. It can be prevented by:
(a) adding a chelating agent (b) adding an antioxidant
(c) protecting it from light (d) suppressing its solubility

7. The storage directions of parenteral solution specify 'store in a cool place'. This may be stored in:
(a) an air-conditioned area at 10°C (b) a refrigerator at 15°C
(c) a place whose temperature is set at 5°C (d) room temperature at 25°C

8. Which term describes the closeness of two or more measurements to each other, regardless of accuracy?
(a) Accuracy (b) Precision
(c) Span (d) Drift

9. The indicator used in the Mohr method is–
(a) Fe^{3+} (b) Fluorescein
(c) Potassium chromate (d) Eosin

10. Which of the following is the function of the Flame or Emission system in Atomic Absorption Spectroscopy?
(a) To split the beam into two (b) To break the steady light into pulsating light
(c) To filter unwanted components (d) To reduce the sample into atomic state

11. Which type of filtration is preferred in gravimetric analysis?
(a) Cold filtration (b) Hot filtration
(c) Vacuum filtration (d) Centrifugation

12. What does the term *'retention time'* refer to in the context of HPLC analysis?

(a) Time taken for sample injection
(b) Time taken for a compound to travel through the column
(c) Time taken for column equilibration
(d) Time taken for detector response

13. G proteins are made up of:

(a) alpha, beta and gamma subunits
(b) alpha and beta subunits
(c) alpha subunits only
(d) alpha, beta, gamma and delta subunits

14. Idiosyncratic reaction of a drug is:

(a) A type of hypersensitivity reaction
(b) A type of drug antagonism
(c) Unpredictable, inherent, qualitatively abnormal reaction to a drug
(d) Quantitatively exaggerated response

15. Anaphylaxis is a severe, life-threatening allergic reaction that is classified as a:

(a) Type A ADR
(b) Type B ADR
(c) Type C ADR
(d) Type D ADR

16. The absorption of tetracycline is significantly reduced when co-administered with antacids or dairy products due to:

(a) Increased gastric motility
(b) Formation of insoluble chelates
(c) Alteration of gut flora
(d) Inhibition of P-glycoprotein

17. Allopurinol increases the activity and toxicity of azathioprine and 6-mercaptopurine by inhibiting which enzyme?

(a) CYP2C9
(b) Thiopurine methyltransferase (TPMT)
(c) Xanthine oxidase
(d) Aldehyde dehydrogenase

18. What is the initial step in drug discovery?

(a) Filing a new drug application
(b) Identifying disease symptoms
(c) Target identification and validation
(d) Drug formulation

19. Vanishing creams are _____ emulsions.

(a) W/O emulsions
(b) Multiple emulsions
(c) Not emulsions
(d) O/W emulsions

20. Endotoxins are a subset of pyrogens that come from _____.

(a) Gram-negative bacteria
(b) Gram-positive bacteria
(c) Coronavirus
(d) Mycotoxins

21. Treated soda lime glass for parenteral packaging is classified as a _____.

(a) Type I glass
(b) Type II glass
(c) Type III glass
(d) Type IV glass

22. When a small amount of a potent substance is to be mixed with a large amount of diluent, the _____ method is used to ensure the uniform distribution of the potent drug.

(a) Levigation
(b) Trituration
(c) Spatulation
(d) Geometric dilution

23. A pH of 1 is _____ times more acidic than a pH of 2.

(a) 2 (b) 100

(c) 10 (d) 4

24. Which of the following is a type of process validation?

(a) Design Qualification (DQ) (b) Operational Qualification (OQ)

(c) Installation Qualification (IQ) (d) Prospective validation

25. Non-polar solvents dissolve non-polar solutes through _____.

(a) induced dipole interactions (b) their high dielectric constants

(c) acid-base reactions (d) hydrogen-bond formations

26. Drugs having ester and amide labile groups in their molecular structure degraded through _____.

(a) oxidation in the presence of air (b) hydrolysis in the presence of water

(c) photolysis in the presence of visible light (d) general acid-base catalysis

27. For pharmaceuticals, Shelf-life (Expiry date) is usually calculated as the time to reduce to _____ of the initial amount.

(a) 80% (b) 90%

(c) 50% (d) 60%

28. BCS Class II drugs are:

(a) highly soluble, highly permeable (b) poorly soluble, poorly permeable

(c) highly soluble, poorly permeable (d) poorly soluble, highly permeable

29. The AUC of a diclofenac sodium from a 37.5 mg IV dose was 150 $mcg.ml^{-1}.hr$. When the same dose was given orally, the AUC was found to be 75 $mcg.ml^{-1}.hr$. The absolute bioavailability will be _____.

(a) 50% (b) 2%

(c) 75% (d) 18.75%

30. The wash out period during cross-over bioequivalence study design is usually about _____ times the half-life of the drug.

(a) three (b) five

(c) ten (d) twelve

31. A solid dispersion technique is used to _____ of poorly soluble drugs.

(a) reduce the particle size (b) increase the contact angle

(c) reduce the bioavailability (d) improve the dissolution rates

32. A clear, sweetened hydroalcoholic solution intended for oral use and is usually flavoured to enhance its palatability is called _____.

(a) Syrup (b) Elixir

(c) Liniment (d) Tincture

33. Consider the following statements on *Water for Injection (WFI)*:

I. The total solid contents of WFI should not be more than 1mg/100 mL of WFI.

II. Although water for injection is not required to be sterile, it must be pyrogen-free.

(a) Only I is true (b) Only II is true

(c) Both I and II are false (d) Both I and II are true

34. In *parallel line* bioassay, the minimal design is the 2+2 assay, in which _____.

(a) two doses of standard and two doses of unknown are used

(b) two standards and two unknowns are used

(c) two animals are used for standards and unknowns

(d) only four responses are recorded

35. The appropriate statistical test for analysing quantal bioassay data would be _____.

(a) probit analysis (b) student t-test

(c) linear regression analysis (d) ANOVA test

36. The most abundant and extensively studied phase I drug-metabolizing enzymes are _

(a) Cytochrome P-450 isozymes (b) Dihydropyrimidine dehydrogenases

(c) Thiopurine S-methyltransferases (d) N-Acetyltransferases

37. An appropriate term conventionally used to describe a more gradual decrease in responsiveness of a receptor to a drug, taking days or weeks to develop is _____.

(a) Tachyphylaxis (b) Drug resistance

(c) Tolerance (d) Antagonism

38. Therapeutic index is calculated as:

(a) ED_{50}/LD_{50} (b) E_{max}/ED_{50}

(c) LD_{50}/ED_{50} (d) E_{max}/LD_{50}

39. Which one of the following examples is not a pharmacodynamic interaction?

(a) β-Adrenoceptor antagonists diminish the effectiveness of β-adrenoceptor agonists such as salbutamol

(b) Warfarin competes with vitamin K, preventing hepatic synthesis of various coagulation factors

(c) Monoamine oxidase inhibitors increase the amount of noradrenaline stored in noradrenergic nerve terminals and interact dangerously with drugs, such as ephedrine

(d) Gastrointestinal absorption is slowed by drugs that inhibit gastric emptying, such as atropine or opiates

40. Toxicity testing, pharmacokinetic analysis and formulation are performed during _____.

(a) Drug discovery stage (b) Preclinical development stage

(c) Clinical development stage (d) Post-marketing surveillance stage

41. The most widely used *in vitro* tests for mutagenicity is _____.

(a) GFR test (b) Gastruloids test

(c) Comet Assay (d) Ames test

42. According to ICH Guideline on impurities (Q3C), Class 1 solvents are:

(a) Solvents with low toxic potential (b) Solvents to be limited

(c) Solvents to be avoided (d) Solvents with strong odour

43. The most commonly used chelating agent in complexometric titrations, EDTA is a _____.

(a) bidentate ligand (b) tridentate ligand

(c) hexadentate ligand (d) tetradentate ligand

44. The purpose of digestion in gravimetry is _____.

(a) adjustment of the pH of the solution (b) to make larger crystals

(c) to remove the moisture (d) to remove the impurities

45. According to the Beer-Lambert's law, absorbance is directly proportional to the _____.

(a) concentration only
(b) path length only
(c) concentration and path length
(d) wavelength only

46. In gas chromatography, _____ remains the most commonly used carrier gas in about 90% of instruments.

(a) Helium
(b) Argon
(c) Nitrogen
(d) Hydrogen

47. This GC detector is also known as Katharometer.

(a) Flame Ionization Detector
(b) Thermal Conductivity Detector
(c) Electron Capture Detector
(d) Flame Photometric Detector

48. Which HPLC detector is used in the analysis of carbohydrates?

(a) UV-Vis detector
(b) Light scattering detector
(c) Refractive index detector
(d) Fluorescence detector

49. For normal phase HPLC, the mobile phase is _____.

(a) non-polar
(b) polar
(c) ionic
(d) surfactants

50. To determine the Limit of Detection (LOD), the signal-to-noise ratio of the peak of analyte of interest in the sample should be at _____.

(a) 10:1
(b) 3:1
(c) 1:10
(d) 1:3

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